

Synthetic Market Phase 14

Affordance-ladder geometry on the same 4-asset market frame used in Phases 9-13. This phase asks whether progressively stronger execution constraints deform the shared market coordinates like risk and portfolio, or whether affordance behaves more like route masking.

DATE	BASE DATA	STATES	OBJECT
22 March 2026	Phase 14 affordance ladder	row_mean @ L1, row_eos @ L25	adjacent affordance-step transforms

MAIN READ

The shared 4-asset market frame survives the full affordance ladder almost perfectly early, but late states look sharper and more route-masking than either risk or portfolio.

Early row_mean states keep both latent axes above 0.990 R^2 all the way to affordance_5, so the base market coordinates are still present. Late row_eos states do move toward affordance-adjusted score geometry, but the best adjacent fits favor flexible local maps and the full market_only -> affordance_5 warp is only moderate. The strongest interpretation is preserved shared geometry plus increasingly strong route masking.

EARLY FRAME FLOOR	LATE STRONGEST LOCAL	LATE END-TO-END	NONTRIVIAL COMPOSE
0.990 R^2	0.881 R^2	0.426 R^2	0.924 cos
market_only -> affordance_5, latent_y	linear fit on affordance_2 -> affordance_3	best direct family on market_only -> affordance_5	late linear composed-vs-direct matrix cosine

Why Phase 14 Exists

Phases 12 and 13 established two complementary context stories on the same 4-asset object:

- risk preserved a shared market frame and behaved like a globally coherent near-rigid ladder
- portfolio preserved the same frame, but looked more local and redistributive

Affordance is the next useful family because it changes *execution availability* rather than preference weight. In the synthetic prompts:

- the strongest asset is capped first
- the second asset becomes partially restricted and then blocked

- the third asset becomes size-limited late
- by `affordance_5`, only the weakest route remains fully open

So this phase asks whether the model treats execution constraints as:

- another smooth deformation of the shared market frame
- or a sharper route-masking process layered on top of that frame

Experimental Design

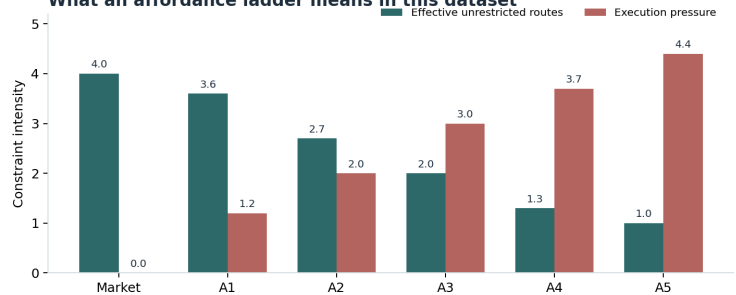
Phase 14 dataset design and experimental target

What data was used

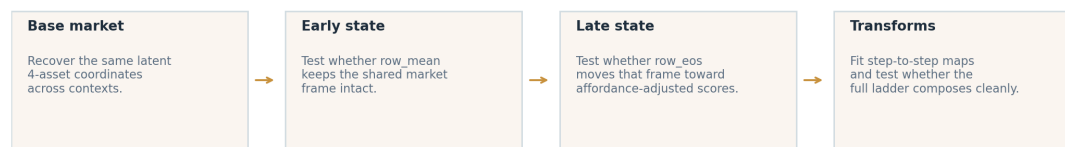
The dataset repeats the same 4-asset markets under progressively harder execution constraints.

1. 4 latent market scenarios
 2. 2 surface styles
 3. 4 row/symbol permutations
 4. 3 global magnitude scales
 5. 6 affordance contexts
- Total prompts: 576
Asset rows: 2304
Pairwise rows: 6912

What an affordance ladder means in this dataset



What is being tested



Phase 14 is built from repeated variants of the same 4-asset markets. The figure shows the dataset recipe, what the affordance ladder means, and the actual question being tested.

The easiest way to read this phase is:

- start with a base 4-asset market
- render many nuisance variants of the same market
- place each variant into six contexts: `market_only`, then `affordance_1`..`affordance_5`
- ask whether the model keeps the same market coordinates and how those coordinates change across the ladder

Concretely, the dataset contains:

- 4 latent market scenarios
- 2 surface styles
- 4 row / symbol permutations
- 3 global scale variants
- 6 contexts per base market

That yields:

- 576 prompts total
- 2,304 asset rows
- 6,912 pairwise asset comparisons

In this report, the surface styles, row permutations, and global magnitude scales are all *nuisance variants*. That means they change how the same market is written on the page without changing the underlying latent market shape we actually care about.

What “Affordance Ladder” Means

In this report, a “ladder” just means an ordered sequence of matched prompt variants. The market stays the same, but the contextual pressure changes step by step.

For Phase 14:

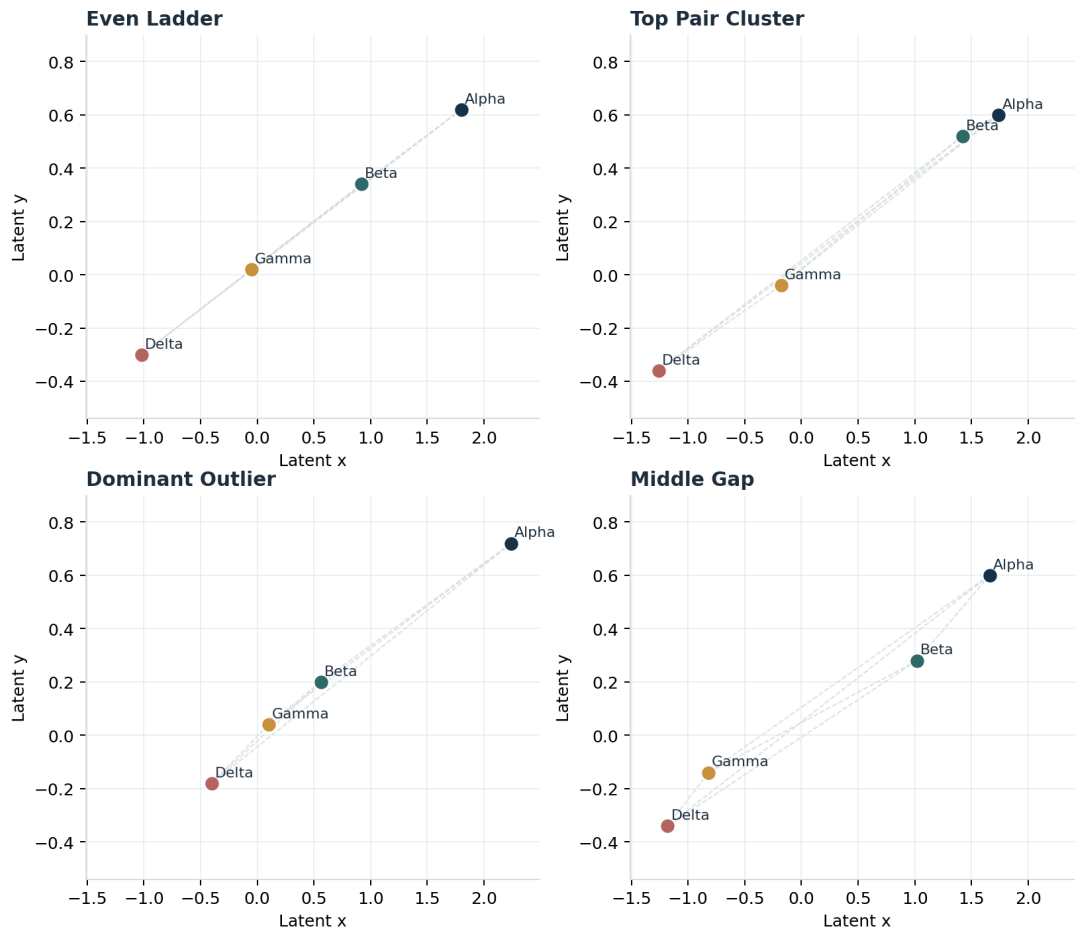
- market_only has no hard execution constraints
- affordance_1 lightly caps the current leader
- affordance_3 blocks the top two routes
- affordance_5 leaves only the weakest-ranked route fully unrestricted

So the ladder is not a time series. It is a controlled constraint sweep over the same base market.

Context	Constraint severity	Instructional effect
market_only	none	No hard execution constraints are supplied.
affordance_1	light	The strongest route is capped for new adds.
affordance_2	light-medium	The strongest route is capped and the next route becomes confirmation-only.
affordance_3	medium	The top two routes face hard execution limits.
affordance_4	strong	Top two blocked; third route can only be added in small size.
affordance_5	strongest	Only the weakest-ranked route remains broadly open.

What “4-Asset Market Geometry” Means

The 4-asset geometry object: same assets, different whole-market shapes



Each panel is a different latent market shape. The goal is not just to recover a winner, but to preserve the relative placement of all four assets at once.

The four latent market-shape families used in the dataset. Each point is one asset. Dashed edges mark the pairwise distances that collectively define the whole-market geometry.

The key object here is not a single asset row and not a single pairwise comparison. It is the *shape of the 4-asset market*.

That means:

- four assets are placed at known latent coordinates
- those coordinates define six pairwise distances
- the distances together determine the market’s overall shape

Why this matters:

- two markets can have the same winner but different shapes
- two markets can have the same rank order but different spacing or clustering
- a real set-level representation should preserve more than “which token looks best”

So when the report says “shared market frame,” it means a reusable coordinate system that organizes all four assets together.

Scenario name	Plain-language meaning
even ladder	The four assets step down in fairly regular intervals. There is a clear ordering, but no special clustering.
top pair cluster	The top two assets sit close together while the lower two are farther away.
dominant outlier	One asset is clearly separated above the other three, which are comparatively compressed.
middle gap	There is a larger-than-usual split between the upper pair and lower pair of assets.

Plain-Language Terminology

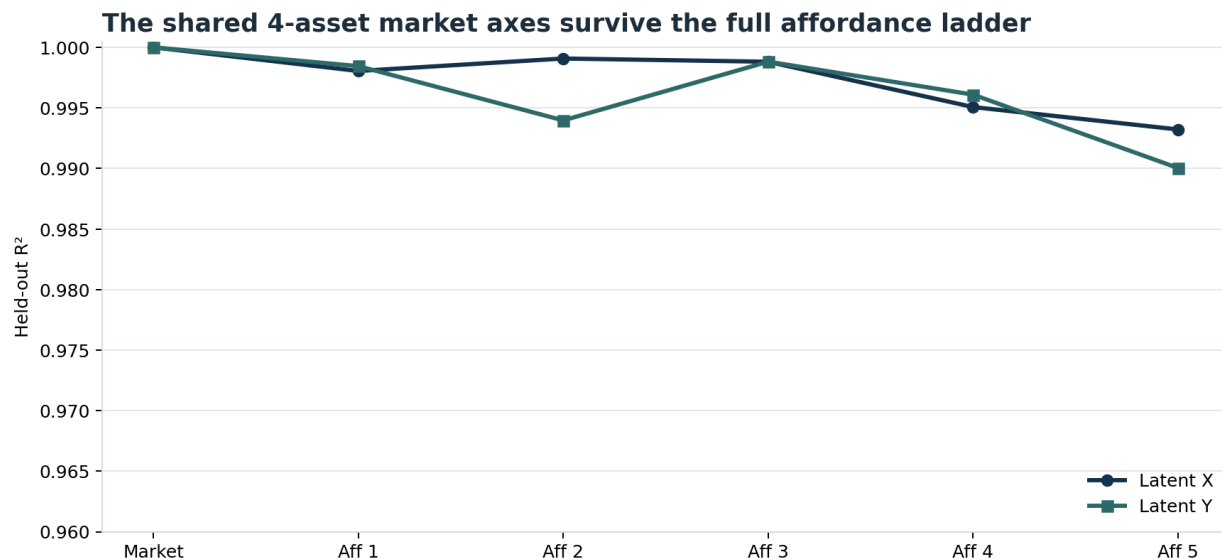
This is the shortest way to decode the jargon used later in the report:

- market frame the shared coordinate system the model seems to use to organize all four assets
- market coordinates the x/y position of each individual asset inside that shared frame
- market geometry the whole shape formed by those four coordinates together
- rigid map a transform that mostly keeps distances and angles intact, like a rotation or reflection
- context-adjusted score geometry the asset layout implied after we rescore the same market under a specific context, such as progressively stronger execution constraints
- nuisance variants prompt differences that should not change the underlying market meaning, like row order, symbol aliases, or formatting style

The point of Phase 14 is to separate:

- real geometry changes caused by context
- from fake changes caused by nuisance presentation differences

The Shared Market Frame Still Survives the Full Ladder



Held-out coordinate transfer from `market_only` into each affordance context. Both latent axes stay high if the shared frame survives.

This is the strongest stability result in the report. A probe trained on `market_only` coordinates still recovers the same latent axes in the affordance contexts with very little loss:

- `market_only` -> `affordance_1`: 0.998 / 0.998 R² on x / y
- `market_only` -> `affordance_3`: 0.999 / 0.999 R²
- `market_only` -> `affordance_5`: 0.993 / 0.990 R²

The best layers remain extremely early: `row_mean` @ L1 for most transfers, with only the y axis in `affordance_4` and `affordance_5` shifting slightly to L4.

So affordance does *not* erase the base market organization. Even when the strongest routes are progressively capped or blocked, the model still keeps a reusable 4-asset coordinate system that looks almost identical to the original market frame.

Late States Still Realign Toward Affordance-Adjusted Score Geometry

Later row_eos states shift toward affordance-adjusted score geometry across the ladder



Realignment margin between recovered activation-space geometry and affordance-adjusted score geometry for each context.

Late states do not stay purely latent. The best realignment margins all occur on row_eos, which means the end-of-row state is moving toward the affordance-adjusted score layout rather than staying at the raw latent market geometry.

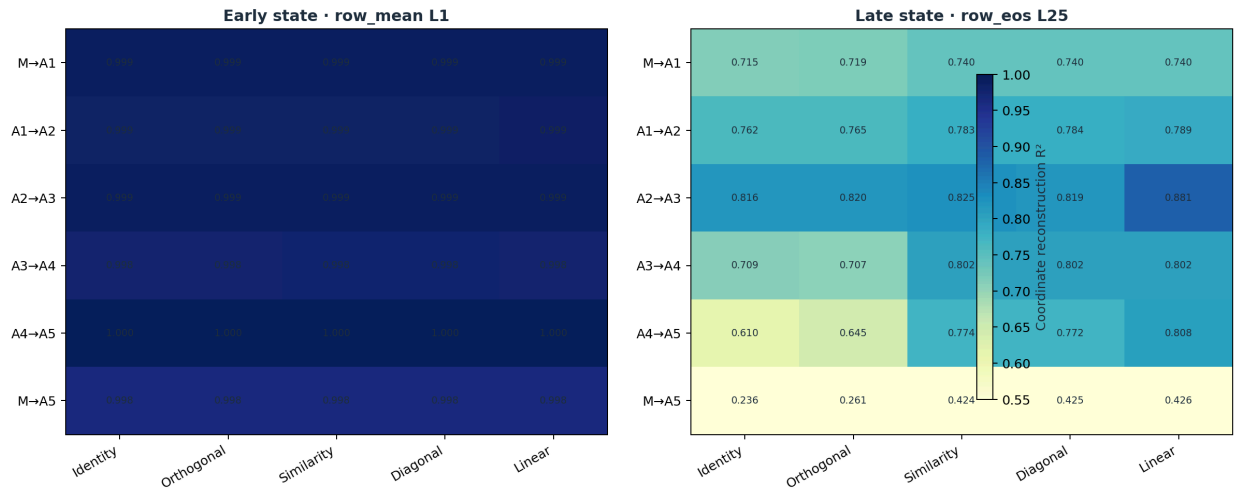
The pattern is uneven:

- market_only: margin +0.024 at L14
- affordance_1: +0.029 at L31
- affordance_2: only +0.012 at L42
- affordance_4: strongest margin, +0.072 at L36
- affordance_5: still strong, +0.049 at L25

So the affordance ladder does reshape the late geometry, but it does so most clearly once the restrictions become genuinely severe. Mild execution caps are comparatively weak.

Early Affordance Transforms Are Nearly Trivial

Affordance-step transforms reveal which local maps stay near-rigid and which do not



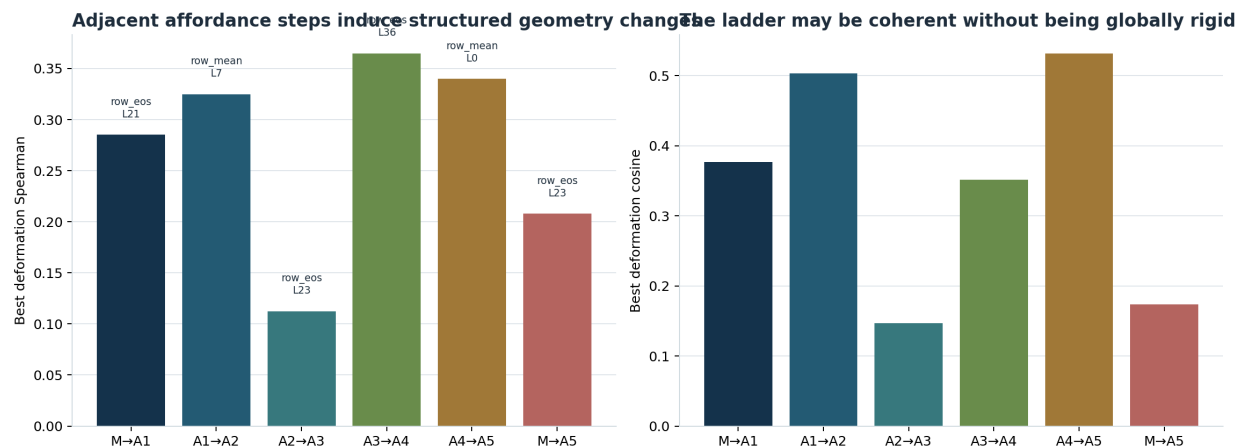
Coordinate reconstruction R^2 for transform families fit on each adjacent affordance step. Left: early state. Right: late state.

The early state is almost too clean. Every adjacent affordance step is already extremely well described by a trivial family:

- market_only -> affordance_1: best diagonal, 0.9993 R^2
- affordance_1 -> affordance_2: best linear, 0.9989 R^2
- affordance_2 -> affordance_3: best linear, 0.9994 R^2
- affordance_3 -> affordance_4: best similarity, 0.9984 R^2
- affordance_4 -> affordance_5: best linear, 0.9998 R^2
- full market_only -> affordance_5: best identity, 0.9976 R^2

That means the early recovered frame barely changes at all. The model is still carrying an almost unchanged market layout before the later context-sensitive reshaping begins.

The Late Ladder Is Local, Mixed, or Mask-Like



Best deformation margins for adjacent and end-to-end affordance comparisons. Higher margins indicate stronger evidence that the geometry is shifting toward the context-adjusted arrangement.

This is where affordance starts to separate from risk and portfolio.

On the late state, the strongest adjacent-step fits are all local and flexible:

- affordance_1 -> affordance_2: best linear, 0.789 R^2
- affordance_2 -> affordance_3: best linear, 0.881 R^2
- affordance_3 -> affordance_4: best linear, 0.802 R^2
- affordance_4 -> affordance_5: best linear, 0.808 R^2

The deformation margins tell the same story. The largest positive shifts are not end-to-end:

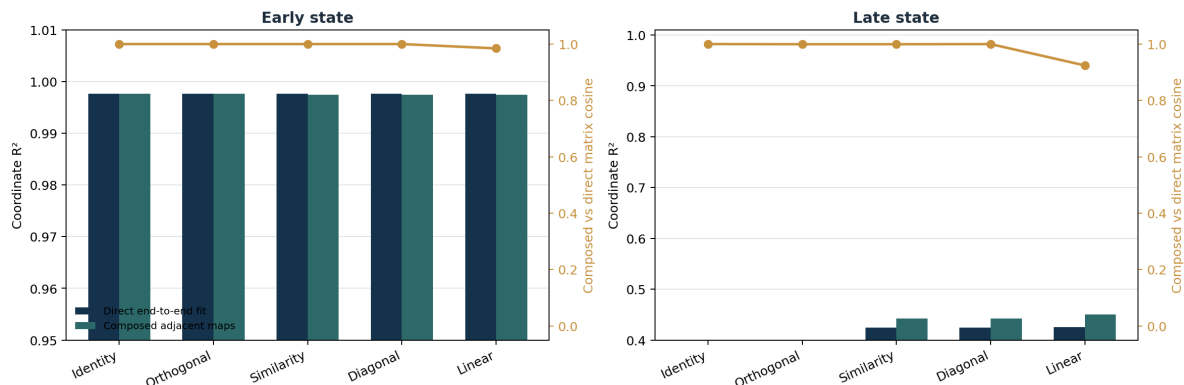
- affordance_3 -> affordance_4: +0.365
- affordance_4 -> affordance_5: +0.340
- affordance_1 -> affordance_2: +0.325

By contrast, the direct end-to-end market_only -> affordance_5 deformation is only +0.208, and its best direct transform fit is just 0.426 R^2 .

So late affordance behavior looks less like one smooth global warp and more like a sequence of sharper local reallocations as routes get partially blocked and then removed.

Composition Keeps Orientation Coherent, or Breaks It

End-to-end affordance behavior can be compared directly against the composition of local steps



For market_only -> affordance_5, compare direct end-to-end fits against the composition of all adjacent affordance-step maps. Gold marks matrix cosine between the composed and direct transforms.

The composition result is mixed in a revealing way.

For the late market_only -> affordance_5 comparison:

- direct identity: 0.236 R^2
- direct orthogonal: 0.261 R^2
- direct similarity: 0.424 R^2
- direct diagonal: 0.425 R^2
- direct linear: 0.426 R^2

If we compose the adjacent-step maps instead of fitting the end-to-end step directly, the flexible families improve slightly:

- composed similarity: 0.443 R^2
- composed diagonal: 0.442 R^2
- composed linear: 0.451 R^2

But the matrix-cosine story is still constrained:

- orthogonal: 0.9994
- similarity: 0.9994
- diagonal: 0.9999
- linear: 0.9242

So the ladder keeps a broadly coherent orientation, but the late end-to-end geometry is not captured by a single clean rigid map. The best affordance story is a preserved base frame with increasingly strong local route-masking deformations layered on top.

Interpretation

SUPPORTED NOW

Affordance preserves the same shared 4-asset market frame very strongly in early states. Late states do move toward context-adjusted score geometry, especially once the strongest routes are heavily constrained. The best late adjacent-step fits prefer flexible linear transforms, which is consistent with local route closures or selective masking rather than simple preference reweighting.

NOT SUPPORTED

The full affordance ladder is not a single smooth global rigid map. Unlike risk, the late `market_only` $\rightarrow$ `affordance_5` step is only moderately recoverable under the best family, and unlike the cleanest risk result, a single end-to-end orthogonal or identity map is not enough to explain the late geometry.

Putting Phases 11-14 together now gives a cleaner taxonomy:

- risk behaves like a globally coherent near-rigid ladder
- portfolio behaves like local redistributive movement inside the same frame
- affordance behaves like preserved frame plus sharper local route masking

That is a stronger and more interpretable result than a generic “context matters” claim. Different context families appear to deform the same underlying market coordinates in qualitatively different ways.

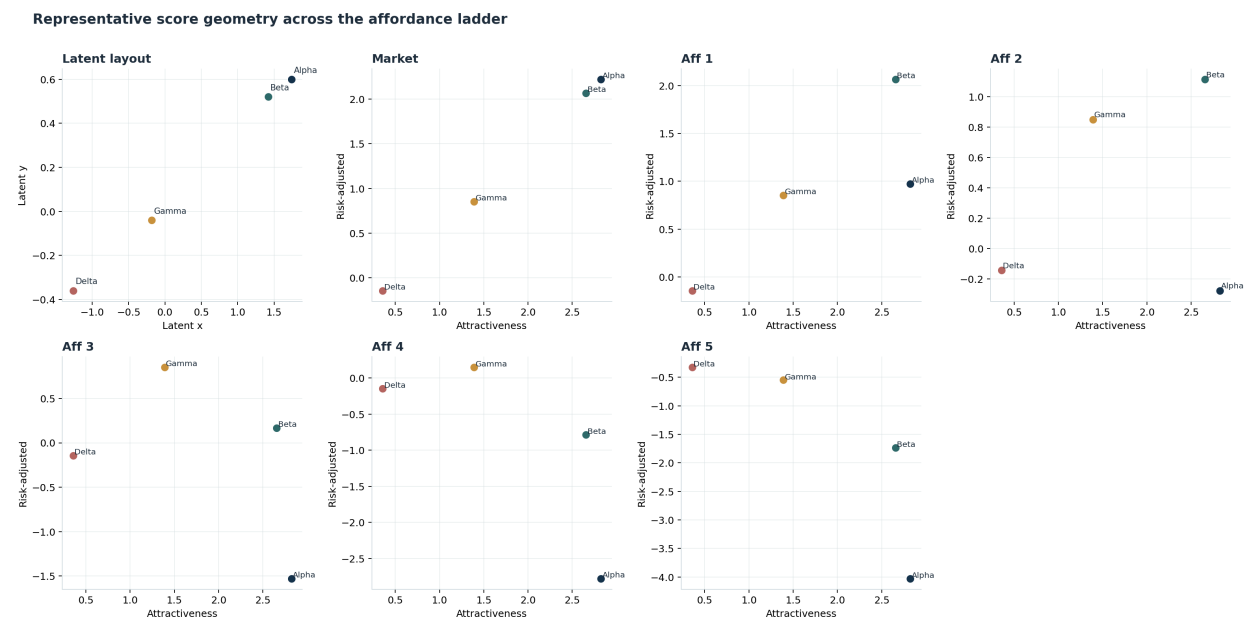
What To Do Next

If Phase 14 keeps the shared frame but looks sharper and less globally coherent than both risk and portfolio, the next step should be a small real-DX bridge using matched prompt variants and the same coordinate-and-deformation lens.

The key question is no longer “does a latent space exist?” The key question is:

- do real execution overlays also look like structured deformations of a shared market frame
- and if so, which context families are smooth, local, or mask-like

Appendix A: Representative Affordance Geometry



Representative top_pair_cluster market from the affordance ladder. The figure shows the same latent base market with affordance-adjusted score geometries layered on top of it.

This figure anchors what “affordance deformation” means in the report. The transform analyses above are attempts to recover how the same underlying 4-asset market is reweighted once the strongest execution paths are progressively capped or blocked.

Appendix B: Glossary

Term	Meaning
shared market frame	The recovered 2D coordinate system that organizes all four assets before context-specific reweighting.
market coordinates	The individual x/y positions of the four assets inside the shared market frame.
market geometry	The whole-market arrangement of those coordinates, including spacing, clustering, and pairwise distances.

Term	Meaning
4-asset geometry	The full relative placement of all four assets at once. In practice this means the coordinate layout or, equivalently, the pattern of six pairwise distances between assets.
affordance ladder	A matched sequence of context variants where the base market is fixed and execution availability is stepped from <code>market_only</code> to <code>affordance_5</code> .
nuisance variants	Prompt changes that alter presentation but should not alter the underlying market meaning, such as row order, formatting style, symbol aliases, or global scale formatting.
coordinate transfer	How well a probe trained in one context can recover the same latent coordinate in another context. High R^2 means the base frame survived.
realignment margin	How much closer the recovered geometry is to the context-adjusted score geometry than to the raw latent layout. Positive margins mean the context is actively reweighting the market.
context-adjusted score geometry	The geometry implied by rescoreing the same four assets under the current context. In Phase 14, that means incorporating progressively stronger execution caps and blocks.
identity	No transform at all. If identity already works, the two contexts share almost the same geometry in the decoded frame.
orthogonal	A pure rotation or reflection. Distances and angles are preserved; only orientation changes.
rigid map	A transform that preserves shape up to orientation. In practice here, the closest approximation is an orthogonal map or a near-rigid similarity map.
similarity	Uniform scaling plus rotation. This captures isotropic compression or expansion.
diagonal	Axis-aligned rescaling in the shared coordinate frame. This captures simple non-uniform compression without rotation.
linear	Any 2×2 linear map. This is the most flexible family here and can absorb rotation, anisotropic scaling, and shear-like behavior.
matrix cosine	Cosine similarity between two flattened transform matrices. Here it is used to compare direct end-to-end maps against the composition of adjacent ladder steps.